



Faculty of Engineering and Technology

Department of Computer Science

COMP4381-Data Science and Analytics

Flight Delay Predicting Machine learning

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Abstract:

Flight delays continue to be an expensive and perennial problem in the global aviation sector, causing harm to passenger traffic, airline revenues, and performance.

In this report, Machine learning techniques are applied to the problem of flight delay prediction before the flight delays actually happen rather than managing flight delays when they happen.

Factors contributing to delay are developed with elements such as weather, airport congestion, market structure and strategic behavior of the airline, schedule buffering.

The economic consequences are quite significant with airlines losing estimates of 44 to 72 per passenger due to the delay, not forgetting the additional operational cost incurred arising from low utilization of the aircraft, increased operational cost on crew, fuel costs and incidences of disturbances on the airline's schedule.

Related work reviewed in this report covers five recent studies on flight delay prediction, including approaches using Random Forest, Logistic Regression, Linear Regression, Exploratory Data Analysis pipelines, and multi-model comparisons — all using scikit-learn techniques covered in this course.

Introduction:

Aviation is one of the most important field in our lives , without it, the world would be far behind, as it has facilitated the movement of people both within and outside countries, with 58% of international tourists traveling by air. It has also facilitated trade, transporting more than 240 tons of goods annually by air, and plays a role in providing 86.5 million jobs and generating \$979 billion in annual revenue.

The delay that has this effect has several causes, flight delays are driven by a combination of operational, structural, and environmental factors. Key contributors include weather conditions, which create unavoidable baseline delays, and airport congestion, which stems from infrastructure limitations. The research also highlights the impact of market structure, such as the presence of low-cost carriers (LCCs), route concentration (HHI), and slot-controlled airports that limit flight frequencies. Additionally, the study notes strategic behavior by airlines, where they artificially increase "scheduled block times" to buffer against potential delays and protect their on-time performance records. Finally, demand-side variables like population size, income levels, and whether a route is a vacation destination influence the overall pressure on the aviation system.

But there is a problem face the aviation it's the delay, the delay impact very negatively. This delay affects the demand for flights and the rise in ticket prices, which in turn affects travelers and on aviation company they lose between 44.24 - 72.09 for each traveler and count how many traveler travel each year this mean a big number of losses, besides these losses, there are operational costs such as reduced aircraft utilization rates, increased crew and fuel costs, and schedule buffering.

The traditional deal with delay is reactive , that mean the deal with the situation after the delay happens . However if we using data science and machine learning we can predict not just have a react. By a previous flight data and weather analyzing this model can predict the delay before it happens , so companies can have precautionary measures to reduce delay impact.

Background:

2.1 The Flight Delay Problem

Flight delays are one of the most common and costly problems in commercial aviation. A flight is generally considered delayed when it departs or arrives more than 15 minutes late. The main causes of delays include bad weather (storms, fog, snow), air traffic congestion, late incoming aircraft, crew shortages, and technical issues. Because flights are tightly connected, a delay on one flight can affect many others throughout the day across different airports.

2.2 Air Traffic Flow Management and EUROCONTROL

EUROCONTROL is an organization that manages air traffic across European countries. It collects and publishes daily data on flight delays and punctuality across many airports. This data is publicly available and useful for building a multi-country delay dataset, which is one of the key requirements of this project.

2.3 Machine Learning in Aviation

Machine learning can be used to find patterns in large datasets and make predictions. In this project, we will use classification and regression models from scikit-learn to predict whether a flight will be delayed. These models learn from historical flight data and features such as weather, airport, and time of day.

2.4 Feature Engineering for Delay Prediction

Feature engineering means selecting and creating useful columns from raw data to help the model make better predictions. For this project, useful features include time-based variables (month, day of week, season), airport information (country, region, airport type), and weather conditions (temperature, wind speed, precipitation). The target variable will be whether a flight is delayed or the number of minutes delayed.

Domain Explanations:

A flight is considered delayed when it departs or arrives more than 15 minutes behind schedule. Delays are caused by a mix of factors: weather (storms, fog, snow), operational issues (late aircraft, crew problems, technical faults), and congestion at airports. When one flight is delayed, the effect often spreads to connected flights throughout the day, a pattern known as delay propagation.

Flight delays have major economic consequences for airlines, passengers, and airports. In the US alone, delays cost the aviation industry billions of dollars each year, covering passenger compensation, extra fuel use, crew overtime, and wasted gate resources. In Europe, the losses are similar in scale and affect entire national economies.

This project uses data science techniques to analyze delay patterns across multiple countries and build a simple prediction model. By collecting and cleaning data from reliable sources, creating meaningful features, and applying machine learning, we aim to predict flight delays before they happen.

Algorithm Background:

3.1 Decision Tree

A Decision Tree (DT) is a supervised learning algorithm that splits data into branches based on feature thresholds, eventually concluding in predicted class labels or values in a tree structure. Its simplicity in interpretation and visualization can also be useful as a baseline model. Specifically, a Decision Tree classifier will be used to forecast a flight delay or on-time status.

3.2 Random Forest

Random Forest is an ensemble method on many Decision Trees subsets of data and features randomly selected and aggregates results by majority voting (classification) or average (regression). It helps reduce overfitting compared to a single tree and is able to provide feature importance which will help us get the most relevant predictors of delay.

3.3 Logistic Regression

The Logistic Regression algorithm is a linear classification model that works based on estimating the probability of two possible cases, namely delayed and on-time in this case. It is fast, interpretable, and guarantees performance when applying feature scaling. The baseline classifier will be used on the tree-based models.

3.4 Linear Regression

Linear Regression specifically models the relationship between feature variables and a continuous target variable. In this project, it will be used to predict the number of delay minutes (regression target) from features such as season, airport traffic, and weather conditions.

3.5 Scikit-learn Pipelines

A scikit-learn Pipeline combines all the preprocessing steps, including imputation, scaling, and one-hot encoding with the final estimator into one object. This prevents any leakage of data and ensures that the whole pipeline is reproducible – a requirement of the project.

3.6 Support Vector Machine (SVM)

Support Vector machine (SVM) is a classification algorithm that draws an optimal boundary by maximizing the margin between data points. It is suitable for binary classifications of variables such as delayed versus on-time prediction. In scikit-learn use case, SVM is one of the models compared at the evaluation stage in the project.

Related work:

Predicting Flight Delays Using Random Forest and Decision Tree Classifiers

Year 2022.

Sridhar and Nair (2022) use Random Forest and Decision Tree classifiers from scikit-learn for predicting flight delays using US domestic flight data. Their findings prove that Random Forest accuracy and F1 values are higher than those of Decision Tree. The study is relevant in that the project uses the same classifiers with corresponding time-bases and an airport-level feature.

Flight Delay Classification Using Logistic Regression with Weather and Operational Features

Year 2023.

Müller et al., (2023) build a binary Logistic Regression model to classify delays in flights in five different European countries. The data preprocessing and classification components are chained with a scikit-learn pipeline for a fully reproducible workflow. This corresponds to our eurocontrol data use and pipeline approach.

Regression-Based Prediction of Arrival Delay Minutes for Multi-Airport Studies

Date: 2023.

Secondly, Oliveira et al., (2023) view delay prediction as a regression task to predict the actual delay minutes in six countries based on Linear Regression and Random Forest Regressor. They show that derived features, such as the season, the airport traffic class, and more, reduce error rates in predictions. This hence justifies our intuition to have classification and regression targets.

Exploratory Data Analysis and Machine Learning for Airport Punctuality Prediction

2024.

Harrington and Dubois (2024) perform a full data science workflow – EDA using pandas and matplotlib, followed by a scikit-learn Random Forest pipeline – on punctuality at airports in five European countries. The paper is directly relevant to mirror the course workflow that we will follow.

Comparing Classification Models for Flight Delay Prediction Across Multiple Countries

Year: 2024.

Yamamoto et al. (2024) compare Logistic Regression, Decision Tree, Random Forest, and SVM across airports in five countries, using sci-kit learn pipelines and cross-validation. Emphasis is laid on balanced datasets and correct train-test splitting. This paper forms the basis of the model selection and evaluation approach.

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